

A Novel Inverter Topology for Compact EV and HEV Drive Systems

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Abstract—This paper describes a new modular inverter topology suitable for compact drive systems in electric road vehicles. The topology is based on dc-side cascading of three-phase two-level inverter bridges, each feeding a set of three-phase windings of a motor. This topology is compared by calculation to the known modular high-frequency (MHF) converter, and is found to have similar losses and semiconductor expenditure requirements. Simulations verify the dynamic and static properties of the converter in a realistic application.

NOMENCLATURE

A. Abbreviations

EV	Electric vehicle
HEV	Hybrid electric vehicle
IM	Induction machine
MHF	Modular high-frequency (converter)
PMSM	Permanent-magnet synchronous machine
SPB	Stacked polyphase bridges (inverter)
VSC	Voltage source converter
ICE	Internal combustion engine

B. Symbols

U_{cap}	Cell capacitor dc voltage
k_a	AC side coupling function
k_d	DC side coupling function
u_a	Stator voltage
u_d	Cell dc terminal voltage
i_a	Stator current
i_d	Battery current

Numerals in indices generally refer to harmonic orders.

I. INTRODUCTION

A conventional electric drive for use in an automotive application comprises a battery, a dc-link capacitor, a two-level inverter and a corresponding three-phase electric machine (commonly of PMSM or IM type). Regardless of the machine type, the inverter should provide a voltage of variable frequency and variable amplitude out of a dc voltage, which nominally amounts to a value between 300–600 V. The dc voltage will also vary around the nominal value in the range depending on the loading conditions and the state of charge of the battery. In some cases, a dc/dc converter is placed between

the battery and the inverter(s) in order to boost and regulate the input voltage. In order to minimize the fuel consumption in a HEV (or maximize the operating range of an EV) the efficiency of each component in the electric drivetrain must be sufficiently high so that the energy potentially available during an electric braking of the vehicle is not wasted into heat. Additionally, in order to minimize cost, fulfilling tight volume constraints and realize an overall competitive solution, the intermittent nature of the operating cycle must be taken into account when dimensioning each component of the electric drive.

From the discussion above, it is evident that the realization of a competitive electric drivetrain is indeed a challenging task for the design engineer. An electric drive formed by the integration of the power electronic inverter and the electric machine into a single unit potentially enables substantial advantages including a reduced amount of cabling as well as weight and EMI reductions [1]–[3]. These benefits make the concept highly interesting for several applications including EV and HEV drivetrains.

This paper proposes a modular converter topology based on dc side cascading of conventional three-phase two-level inverter bridges. The concept is well suited for realizing integrated motor drivetrains in automotive applications. The proposed converter topology is compared to the conventional two-level inverter solution and the relatively recently proposed modular high-frequency converter concept [3]–[5]. The paper is outlined as follows. In Section II, the three converter concepts considered are described and the respective advantages and disadvantages are highlighted. In Section III, a comparison of resulting converter losses of the cascaded topologies in a typical automotive application is presented. A tentative control concept for the proposed topology is described in Section IV along with simulation results validating the operation of the converter by this control. Finally, concluding remarks are made in Section V.

II. INTEGRATED MOTOR DRIVETRAINS FOR EVs/HEVs

A. Two-Level Inverter

For the conventional two-level, three-phase inverter depicted in Fig. 1, the switching frequency of the semiconductor devices

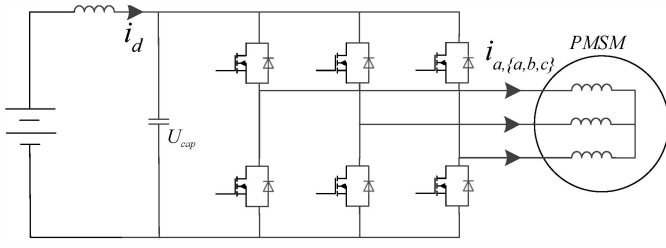


Fig. 1. Two-level inverter feeding a PMSM.

required to handle the dc voltage and ac current (1200 V IGBTs) is limited to below approx 20 kHz. This implies that a fairly large and bulky dc-link capacitor is required. In [6], key properties of dc-link capacitor assemblies in three series produced HEVs are reported confirming that the total capacitor assembly takes up a substantial volume (168–330 cl). While interesting approaches for minimizing capacitor volume requirements are reported in [7], it is evident that the volume requirement of the dc-link capacitor makes the integration of a conventional two-level inverter and a corresponding electric machine challenging [8], [9]. Furthermore, the encapsulation methods used for IGBTs (multi-chip wire-bonded modules) are generally less suited for the type of tight integration required in integral motors. Also, increasing the reliability by implementing redundancy in critical parts cannot easily be made with two-level converters. Attempts at increasing the switching frequency have been made by using soft switching, but this generally adds complexity to the main circuit and reduces reliability since the number of critical points of fault increases [10].

B. Modular High-Frequency Converter

Another development towards increasing reliability and allowing for far-reaching integration of the inverter with the motor was made through the introduction of the modular high-frequency converter (MHF) [3], see Fig. 2. This topology resembles cascaded-cell topologies used for high power applications, such as high voltage direct current (hvdc) power transmission [11], in that it has a number of cells that share the dc voltage between them. Each cell consists of a step-up dc/dc converter (left) and an H-bridge (right), both connected to a common dc link capacitor. The H-bridges of the cells provide single-phase ac voltages which when combined form a symmetric poly-phase voltage for the motor windings. The chopper regulates the power flow from the dc side. Notably, the power flow from each H-bridge will contain a second harmonic fluctuation since the cell ac interface is of single-phase nature. In order to minimize the capacitor size, which is necessary for a compact design, this power fluctuation cannot be absorbed by the capacitor. Since the current on the battery side should be a pure dc current the only way to achieve this is to have the step-up chopper generate an ac voltage at the second harmonic frequency. The governing equations for the steady-state operation of a cell will be described below in order to understand this mechanism in

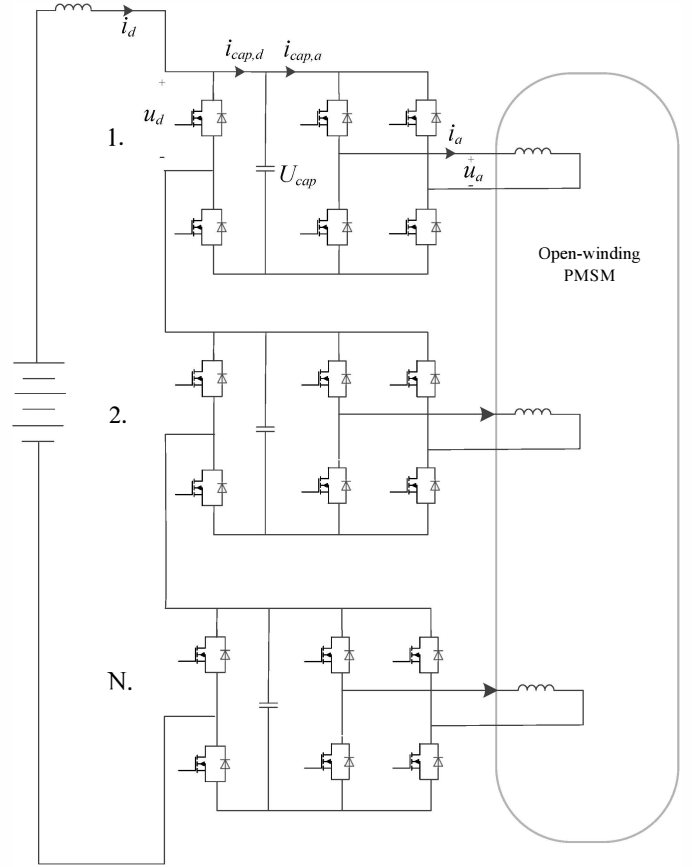


Fig. 2. Modular high-frequency converter feeding PMSM

more detail. Since all expressions relate to a single cell the indices referring to the cell number are dropped for brevity. Firstly, the voltage provided to the motor winding by an H-bridge can be expressed

$$u_a = \hat{k}_{a,1} U_{cap} \cos \omega t \quad (1)$$

where $\hat{k}_{a,1}$ is the fundamental component of the switching function of the H-bridge. High order harmonics in the coupling functions resulting from the pulse-width modulation are neglected for simplicity. Also, to simplify the expressions, the phase displacement of the ac voltage of the concerned cell is set to zero. Accordingly, the current injected into the dc-link by the H-bridge amounts to

$$i_{cap,a} = \frac{1}{2} \hat{k}_{a,1} \hat{i}_{a,1} [\cos(\varphi) + \cos(2\omega t - \varphi)] \quad (2)$$

The corresponding equations for the half-bridge dc/dc converter will be:

$$u_d = [\hat{k}_{d,0} + \hat{k}_{d,2} \cos(2\omega t + \theta_{d,2})] U_{cap} \quad (3)$$

and

$$i_{cap,d} = [\hat{k}_{d,0} + \hat{k}_{d,2} \cos(2\omega t + \theta_{d,2})] I_d \quad (4)$$

Note that, as mentioned, the coupling function contains a second harmonic term (with magnitude $\hat{k}_{d,2}$ and phase $\theta_{d,2}$ to compensate for the power fluctuation from the motor side. The

dc-side 2^{nd} harmonic voltage components from the different cells form a balanced set, and therefore sum to zero in the combined dc-side voltage. The dc power balance in the cell must be upheld and it can be stated

$$\hat{k}_{d,0}I_d = \frac{1}{2}\hat{k}_{a,1}\hat{i}_{a,1}\cos\varphi. \quad (5)$$

Furthermore, as mentioned also the second harmonic currents should be balanced and the relevant equation reads

$$\hat{k}_{d,2}I_d\cos(2\omega t + \theta_{d,2}) - \frac{1}{2}\hat{k}_{a,1}\hat{i}_{a,1}\cos(2\omega t - \varphi) = 0. \quad (6)$$

The only possible solution to this equation is to set

$$\theta_{d,2} = -\varphi \quad (7)$$

and

$$\hat{k}_{d,2}I_d = \frac{1}{2}\hat{k}_{a,1}\hat{i}_{a,1}. \quad (8)$$

If (8) is divided by (5) the following expression is obtained

$$\frac{\hat{k}_{dc,2}}{\hat{k}_{dc,0}} = \frac{1}{\cos\varphi}. \quad (9)$$

It must be fulfilled in order to cancel the 2^{nd} -harmonic current in the dc capacitor. However, the nature of the dc/dc-converter imposes certain limits on the switching function, namely, that $\hat{k}_{d,0} + \hat{k}_{d,2} \leq 1$ and $\hat{k}_{d,2} \leq \hat{k}_{d,0}$. This can be understood by remembering that only instantaneous voltage in the range $[0, U_{cap}]$ can be projected towards the battery side. We can therefore draw two conclusions as regards the operation of the MHF converter. Firstly, the only operating points where it is possible to fully compensate for the 2^{nd} -harmonic ripple is $\varphi = 0$ and $\varphi = \pi$, i.e. pure motoring or pure regenerative operation. If reactive power should be exchanged with the motor a second harmonic ripple will inevitably appear. Secondly, to achieve maximum compensation the dc and 2^{nd} -harmonic components of the switching function should be equal. ($\hat{k}_{d,2} = \hat{k}_{d,0}$). Hence the maximum value of $\hat{k}_{d,0}$ is 0.5, i.e. each cell can take up a fraction of the total dc voltage corresponding to no more than half of the cell dc capacitor voltage.

The attractive properties of the MHF converter consist in that it allows for use of low-voltage unipolar MOSFET devices which have a number of attractive properties. The on-state losses are low compared to high-voltage devices since the on-state resistance (R_{DSon}) generally increases faster than linearly with blocking voltage. Furthermore, using these devices the losses during reverse conduction can be reduced by activating channel, bypassing the internal body diode that otherwise would conduct. This is analogous to the use of such devices for synchronous rectification.

C. Stacked Polyphase Bridges Inverter

In this paper we propose an alternative topology which combines some of the attractive features of the two-level three-phase converter and the MHF, see Fig. 3. It consists in a number of series-connected cells, each consisting of a three-phase bridge with its dc capacitor, and is therefore

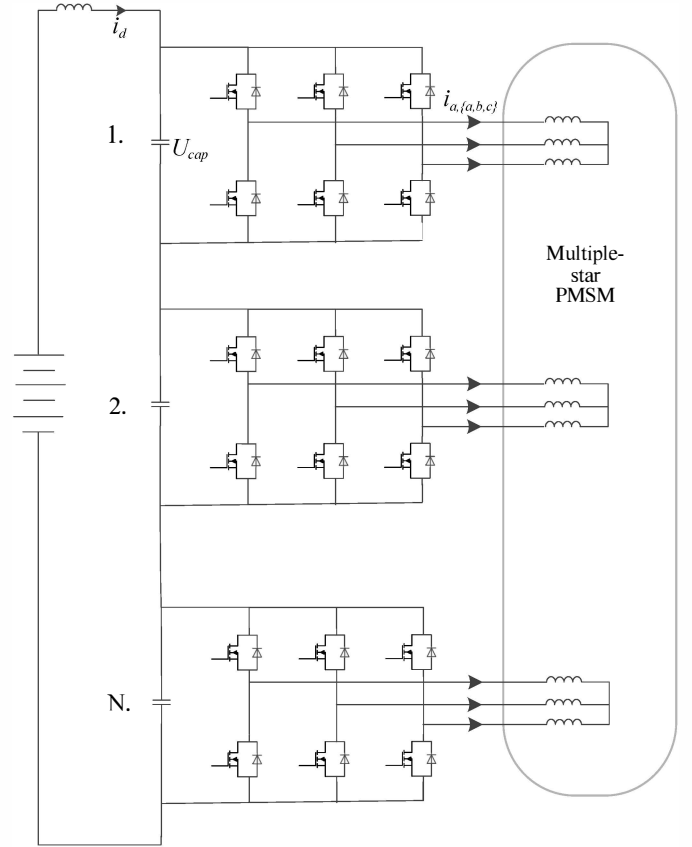


Fig. 3. Stacked polyphase bridges inverter feeding PMSM

labeled *Stacked Polyphase Bridges (SPB) Inverter*. The power conversion has thus only one stage, which could reduce power losses compared to the MHF. The ac terminal of each bridge can provide a symmetric ac voltage, and is in this case connected to one set of three-phase windings of a multi-star PMSM. Similar approaches have been proposed also for high-power ac-fed railway traction systems. The benefits of the topology are mainly that it allows for the use of low voltage semiconductor devices and that the modular structure permits implementation of fault-tolerant operation. As mentioned, the concept has strong similarities with the ordinary two-level inverter. Indeed the topology degenerates into a two-level converter when the number of cells equals one. Since each cell has a three-phase interface to the motor the power flow will be constant, without low frequency fluctuation. As long as no imbalances are present in the three-phase voltages or currents the dc capacitors will only have to absorb high-frequency current harmonics emanating from the pulse width modulation. The control system can also be similar to that of an ordinary inverter. What has to be kept in mind however, is that all of the bridges share the same dc current and that the control has to be adapted accordingly.

III. COMPARATIVE EVALUATION

In this section the SPB inverter is compared to the MHF converter with regard to power losses during operation and

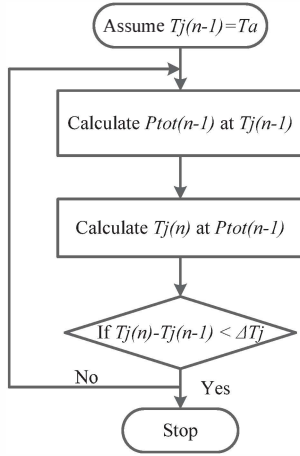


Fig. 4. Loss-thermal iteration model.

TABLE I
PARAMETERS USED FOR COMPARISON

Name	Unit	Value
Rated power	kW	50
Battery voltage, nom.	V	400
Battery voltage, min.	V	350
Battery voltage, max.	V	450
Motor nom. stator frequency	Hz	1000
Switching frequency	kHz	100
Cooling water temp.	C	100

the required silicon area.

Since the power losses and semiconductor temperature depend on each other, loss calculations without a proper thermal model are not accurate. Therefore, this paper uses iteration of thermal and loss models to find correct steady-state losses. Fig. 4 shows the loss-thermal model. Using the instantaneous temperature value derived from the thermal model, temperature dependent parameters can be calculated. Then, the semiconductor loss model uses these parameters to compute the power dissipation. The power dissipation will be delivered to thermal model to calculate a new temperature for the next iteration etc.

A. Reference System Parameters Used in the Comparison

A common reference application is defined in order to allow for a fair comparison of the different concepts. A high-speed integral motor drive system for an electric vehicle is chosen, and the parameters of this system can be found in Table I. Since the switching loss depends on the switching frequency, the comparison is made at the same switching frequency for both converters. This choice should favor the MHF converter since it may require a high switching frequency to minimize battery-side current ripple. It is assumed that a common cooling circuit is used for the electrical drive system and the ICE. Therefore, the cooling water temperature is set to 100 C.

B. Converter Parameters

The converter parameters for the two cases have been chosen so as to ensure a fair comparison with respect to functionality and rating margins of the components. The key parameters for both cases can be found in Table II. The same MOSFETs are chosen for both topologies, namely devices with type number IPB065N15N3G from Infineon, with 150 V blocking voltage, and the parameters were obtained from the datasheet. The silicon area for each valve is varied by connecting an arbitrary number of devices in parallel. As mentioned in Section II B the MHF cells can never project more than half of their capacitor dc voltage towards the battery implying that the overall combined capacitor voltage has to be at least twice the battery voltage. A further 10% margin is added to ensure controllability, resulting in a factor of 2.2 between nominal battery voltage and the sum of the capacitor voltages. Given that the MHF cells have a dc/dc stage as front-end towards the battery a small margin of no more than 50% between the capacitor voltage and the MOSFET blocking voltage has been found to be sufficient. On the other hand, the SPB has no direct means of regulating the cell capacitor voltage other than modifying the torque reference. Since the static and dynamic capacitor voltage control needs further investigation a very wide margin of 120% is set between the nominal cell capacitor voltage and the MOSFET blocking voltage. This way the margin will still be 100% at the upper boundary of the battery voltage range. The number of cells used in each case follows from the above reasoning. The MHF can maintain a constant capacitor voltage and hence also maintain the stator voltage. Therefore, the nominal ac voltage is set to correspond to modulation index 0.9 i.e. 10% from the theoretical limit of 1.0. The same nominal modulation index is used for the SPB although the maximum limit in this case is $2/\sqrt{2} \approx 1.154$ since a zero-sequence component can be used to extend the modulation range. By this choice the SPB can also maintain the nominal stator voltage even when the battery voltage is at its lower limit, and still have a sufficient margin to the maximum modulation boundary. Thus the comparison will also be fair with respect to the capability of delivering constant motor voltage regardless of the battery voltage.

As mentioned in [3], to minimize the MHF converter losses, the distribution of the chip area between the step-up dc/dc converter and the H-bridge should be optimized. In the reported calculations such optimization is made consistently. Also, since the high-side and low-side MOSFETs of the dc-side chopper are slightly unevenly loaded it is beneficial to use different silicon area for them in order to minimize losses given a certain total area [12]. This type of optimization is also used in the calculations of losses for the MHF.

C. Rating and Loss Comparison

Calculated overall semiconductor losses for the two studied converters when operating in the nominal operating point are visualized in Fig. 5. The graph shows the normalized power dissipation as a function of the overall silicon area. As mentioned, the silicon area is varied by varying the total number

TABLE II
CONVERTER PARAMETERS

Name	Unit	MHF	SPB
Cell nom. cap. voltage	V	100	67
Number of cells	-	9	6

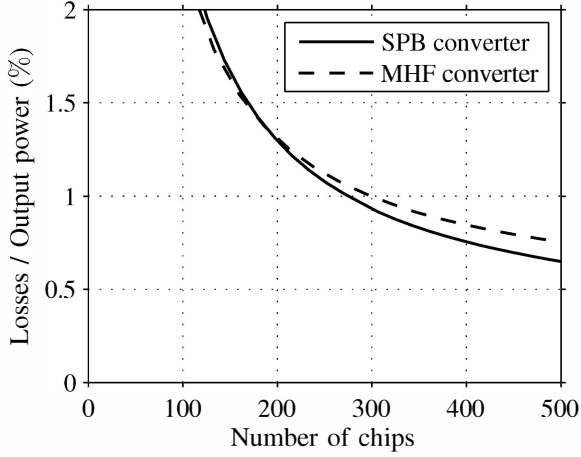


Fig. 5. Comparison of converter losses for different silicon area.

of chips in parallel. For the MHF this is made while the area is distributed in optimised fashion between the different valve functions as discussed previously. It is clear that regardless of the area, the efficiency of the two topologies is similar with the design choices made. In both cases the overall silicon area corresponding to 300 devices of the selected MOSFET will be needed to achieve losses below 1%. Fig. 6 depicts the relationship between junction temperature and the total chip area used for the SPB converter as well as the step-up and H-bridge parts of the MHF converter. From the chart it is obvious that none of the devices will experience excessive junction temperatures for the range of silicon areas studied. It can thus be concluded that for these topologies, where the losses will be spread out between many semiconductor elements, thermal constraints will not be deciding the amount of silicon area. Instead the acceptable loss level will likely be the determining factor. It should be pointed out that the calculations are made under the assumption of liquid cooling with a low thermal resistance from the case to the cooling water.

IV. TENTATIVE CONTROL SCHEME FOR THE STACKED POLYPHASE INVERTER

In this section a brief account is made of a control scheme that was devised for the SPB in order to get an indication of its ability to handle transient conditions. The control system to be used for each cell is essentially the same as that of a conventional PMSM motor drive. It has an inner current control loop and an outer torque control loop. However, care must be taken so that the cell dc capacitor voltage balance is maintained. Since the dc current flowing from the battery is the same for all cells this amounts to controlling the active

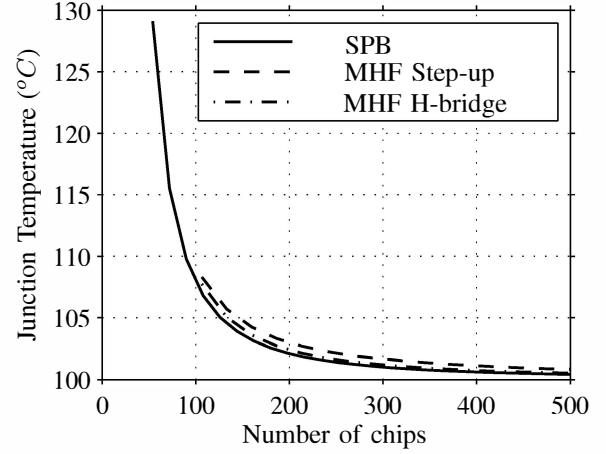


Fig. 6. Comparison of MOSFET junction temperature for different silicon area.

power flow to each set of stator windings. This is made by providing an overall control function accordingly modifying the torque reference for each of the cells, while maintaining the desired total torque [4], [5].

A simulation study has been carried out in order to verify the function of the control strategy. In Fig. 7 simulated waveforms are displayed. The system shown here consists of four cells with varying capacitances. This results in different capacitance voltage ripples and different balancing dynamics. At $t = 0$ ms the system is started and it is evident how the cell capacitor voltages are perfectly balanced within a few milliseconds, although the four cell capacitors were loaded unequally at the starting point. At $t = 1.5$ ms the cell capacitor for one of the cells is shorted, emulating the impact of a cell failure and bypass. After a brief transient the remaining cells take over the voltage and the system enters a new steady-state where the battery voltage is evenly shared between the cells. The simulation study indicates the feasibility of the topology and the scheme for dc capacitor balancing even at fault instances.

V. CONCLUSION

The paper has introduced the SPB inverter as an alternative for highly integrated EV and HEV drivetrains alongside the previously known MHF converter. The common benefits of these cascaded converters compared to conventional topologies are that they allow for the use of low-voltage semiconductor devices and that system reliability can be increased by implementing cell-level redundancy. Against these benefits stands a higher parts count which must be compensated by a lower cost for each component, and possibly less costly manufacturing and assembly methods. The SPB inverter has only one conversion stage, which could imply lower losses. On the other hand the absence of an active front-end towards the battery implies that the cell capacitor voltages can not be directly regulated. The simulations reported in Section IV indicate that

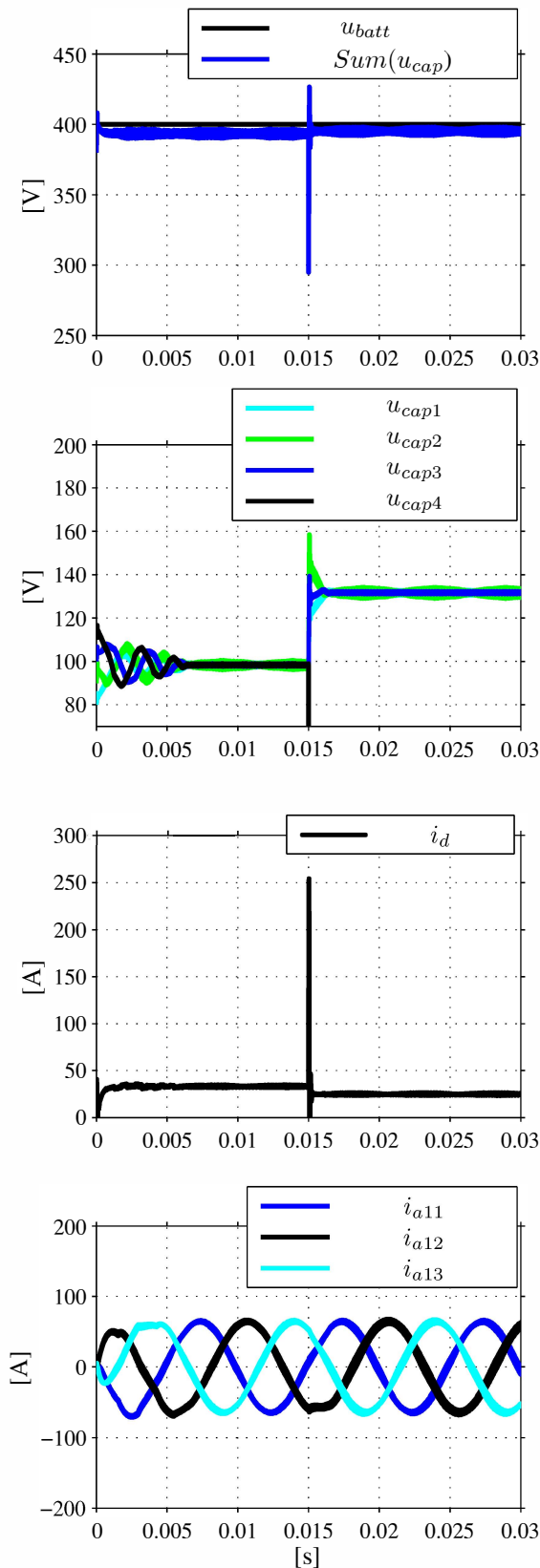


Fig. 7. Simulated waveforms for the SPB inverter. The capacitor in cell 4 is shorted at $t = 1.5$ ms. Chart 1: Battery voltage and sum of cell voltages, Chart 2: Cell voltages, Chart 3: Battery current, Chart 4: Stator currents.

good dynamic and static balancing of the capacitor voltages is possible even with a rather rudimentary approach to the control system design. Still, since the possibilities of capacitor voltage control have not been fully clarified wide design margins have been applied. This way more stacked cells are needed and the conduction losses increase. Still, as shown by the calculations presented in Section III, with these choices the overall semiconductor losses of the two studied topologies are similar, in a typical application, at a given total semiconductor expenditure.

An important issue for further work is the design and analysis of a proper control scheme for capacitor voltage balancing and control under dynamic conditions. This work will establish more firm rules for the requirements concerning the choice of the cell capacitor size and number of cells required for a certain battery voltage.

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